



Tuple Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Sets.

Section 1: Tuple Basics and Creation

★ Create and Display a Tuple

★ Access Elements using Indexing

Negative Indexing in Tuples

★ Tuple Slicing Basics

Single Element Tuple Creation

★ Find Length of a Tuple

Nested Tuple Creation and Access

Tuple of Mixed Data Types

Section 2: Tuple Operations

★ Concatenate Two Tuples

★ Repeat a Tuple N Times

Check Membership using in Operator

★ Count Occurrences of an Element

★ Find Index of an Element

Compare Two Tuples

Convert List to Tuple and Vice Versa

Section 3: Tuple Unpacking

★ Basic Tuple Unpacking

★ Swap Two Variables using Tuple Unpacking

Unpacking with Asterisk (*) Operator

★ Unpack Tuple inside a Loop

Nested Tuple Unpacking

Return Multiple Values from a Function using Tuple

Section 4: Searching and Aggregation

★ Find Maximum and Minimum in a Tuple

★ Sum of All Elements in a Tuple

Average of Elements in a Tuple



Count Even and Odd Numbers in a Tuple

★ **Find Common Elements between Two Tuples**

Sort Elements of a Tuple (using sorted())

Section 5: Tuple-Based Logical Problems

★ **Check if a Tuple is a Palindrome**

★ **Find Duplicate Elements in a Tuple**

Merge Two Tuples Alternately

★ **Convert Tuple of Tuples into a Dictionary**

Find Second Largest Element in a Tuple

Flatten a Nested Tuple

Section 6: Real World Tuple Applications

★ **Student Record using Tuple (Name, Marks, Grade)**

★ **Coordinate Point Storage using Tuple**

RGB Color Value Representation

★ **Employee Record Management using Tuple List**

Date Representation using Tuple (Day, Month, Year)